Clinical Decision Support System for Disease Risk Assessment

Progress Report - 3

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Course: CSIS 4495 – Applied Research Project

Section: 03

Table of Contents

1. Work Date / Hours Log 3
2. Summary Description of Work Done During This Reporting Period 3
3. Repo Check-In of Implementation Completed 4
4. AI Use Section 4
5. APPENDIX 5
   5.1 Appendix A – AI Prompts 5
   5.2 Appendix B – Feature Importance 5
   5.3 Appendix C – Final Heart Failure Model Results 5

1. Work Date / Hours Log

Date	Hours	Description of Work Done
Mar 3, 2026	3.0	Prepared the Heart Failure dataset (heart.csv) for modelling. Encoded categorical variables such as ChestPainType, ExerciseAngina, RestingECG and ST_Slope. Verified dataset structure and feature columns.
Mar 4, 2026	3.5	Conducted exploratory data analysis and generated the correlation heatmap (correlation_heatmap.png) to analyze relationships between clinical indicators and heart disease outcomes.
Mar 7, 2026	3.0	Implemented baseline Logistic Regression and Random Forest models using heart_failure_model.py. Evaluated model performance using accuracy, precision, recall and F1-score metrics.
Mar 8, 2026	4.0	Collaborated with Akif to implement cross-validation and hyperparameter tuning in heart_failure_final_model.py. Generated evaluation artifacts including ROC curves, confusion matrices, and feature importance outputs.

2. Summary Description of Work Done During This Reporting Period

The work done during this reporting period was aimed at creating the heart disease prediction model based on the Heart Failure dataset. The data was ready with the encoding of categorical variables and testing the feature columns needed to run the machine learning algorithms. Exploratory data analysis was also used to analyze the relationship among clinical indicators and a correlation heatmap was created to visualize the relationship.

Logistic Regression and Random Forest were used as baseline classification models and their performance was evaluated using standard performance measures, such as accuracy, precision, recall, and F1-score. These baseline models gave a starting point of how well the models performed on the data.

Further model refinements were done in partnership with Akif such as cross-validation and hyperparameter optimization to enhance model reliability and predictive accuracy. The tuned Random Forest model attained a level of accuracy of about 0.89 which was better than the base models.

The slight problem that was experienced during this phase was the error related to the path of the data sets and file references when loading the data sets to be used in the modelling. To overcome this problem, the project directory layout was sorted out and the path of the dataset files checked. The modelling pipeline is currently operational and can be integrated into the application in the following stage of the project.

3. Repo Check-In of Implementation Completed

During this reporting period, several updates related to the heart disease modelling pipeline were committed to the repository. These updates include model training scripts, evaluation outputs, and visualization artifacts generated during the machine learning workflow.

The following implementation files and artifacts were added or updated:

Model Training Scripts

- heart_failure_final_model.py
- heart_failure_model.py

Datasets

- heart.csv
- heart_cleveland_upload.csv

Model Artifacts

- best_heart_rf_model.pkl
- heart_label_encoders.pkl
- heart_feature_columns.pkl

Evaluation Outputs

- lr_confusion_matrix.png
- rf_confusion_matrix.png
- tuned_rf_confusion_matrix.png
- roc_curve_comparison.png
- tuned_rf_roc_curve.png

Feature Importance Outputs

- rf_feature_importance.csv
- rf_feature_importance.png
- tuned_rf_feature_importance.csv
- tuned_rf_feature_importance.png

Exploratory Data Analysis

- correlation_heatmap.png

These files document the development of the heart disease prediction model and provide visual and analytical outputs used to evaluate model performance.

4. AI Use Section

AI Tool	Version	Purpose of Use	Value Added by User
ChatGPT	GPT-5 (Premium)	Guidance on generating correlation heatmaps and analyzing relationships	Used the guidance to implement the correlation heatmap in Python and interpret relationships between variables such as cholesterol, age,

		between clinical indicators in the dataset	and heart disease outcomes during exploratory data analysis.
ChatGPT	GPT-5 (Premium)	Guidance on saving trained machine learning models for later use	Applied the suggested approach to export the trained Random Forest model and related files using joblib so that the model can be reused during the system integration stage.

5. APPENDIX

5.1 Appendix A – AI Prompts

1. How can I generate a correlation heatmap in Python using seaborn to analyze relationships between variables in a medical dataset
2. How can I save a trained machine learning model in Python so that it can be loaded later in another application

5.2 Appendix B – Feature Importance

Feature importance analysis was performed using the tuned Random Forest model trained on the Heart Failure dataset. The analysis identified **ST_Slope, ChestPainType, MaxHR, Cholesterol, and Oldpeak** as the most influential predictors of heart disease.

The results were visualized using the generated feature importance plot (`tuned_rf_feature_importance.png`). This analysis helps improve the interpretability of the model by identifying which clinical indicators contribute most to prediction outcomes.

5.3 Appendix C – Final Heart Failure Model Results

```
============================================================
FINAL TUNED RANDOM FOREST RESULTS
============================================================
Tuned Random Forest Accuracy: 0.8913043478260869
              precision    recall  f1-score   support

           0       0.90      0.85      0.88        82
           1       0.89      0.92      0.90       102

    accuracy                           0.89       184
   macro avg       0.89      0.89      0.89       184
weighted avg       0.89      0.89      0.89       184
```